

Review Guide

Quiz 9 will be given in class, to evaluate your understanding of the following topics:

(1) Separating Hyperplanes

One way we might approach the task of **binary classification** is to try and find a way to “chop” a p -dimensional feature space into two subspaces, by finding a $(p - 1)$ -dimensional **separating hyperplane** and then predicting the **positive** class for points on one side of this hyperplane and the **negative** class for points on the other side.

Similar to how we began the course by focusing on *linear regression*, we can start here by considering hyperplanes defined with respect to **linear functions** of an observation’s p features: the equation

$$\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} = 0 \quad (1)$$

defines a $(p - 1)$ -dimensional hyperplane which achieves the “chopping” of p -dimensional space, since we can take any observation $\mathbf{x}_i$ we might want to predict a (binary) class for and make the prediction by plugging its features $(x_{i1}, \dots, x_{ip})$ into the left side of Equation 1:

$$\hat{y}_i = \begin{cases} +1 & \text{if } \overbrace{\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}}^{\mathbf{x}_i \text{ lies “above” the hyperplane}} \geq 0 \\ -1 & \text{if } \overbrace{\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}}^{\mathbf{x}_i \text{ lies “below” the hyperplane}} < 0 \end{cases} \quad (2)$$

This approach becomes easier to visualize if we consider a scenario with just **two** features: in this case,

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} = 0 \quad (3)$$

defines a **(one-dimensional) line** within the **two-dimensional feature space**, which you can verify by imagining the first feature X_1 plotted along the x -axis and the second feature X_2 plotted along the y -axis of a two-dimensional plot, then solving Equation 3 for x_{i2} :

$$\begin{aligned} \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} &= 0 \\ \Leftrightarrow \beta_2 x_{i2} &= -\beta_0 - \beta_1 x_{i1} \\ \Leftrightarrow \underbrace{x_{i2}}_{y\text{-axis}} &= \underbrace{\left(-\frac{\beta_1}{\beta_2}\right)}_m \underbrace{x_{i1}}_{x\text{-axis}} + \underbrace{\left(-\frac{\beta_0}{\beta_2}\right)}_b \end{aligned}$$

(2) Max-Margin Classifiers (MMCs)

In most cases, if the observations in our dataset *are* linearly-separable in this way, there are actually *many* (infinitely many!) lines that will achieve this separation. So, what criteria might we use to choose the “**best**” separating hyperplane from among this possibly-infinite collection?

The answer, which forms a core “building block” for both Support Vector Classifiers (SVCs) *and* Support Vector Machines (SVMs), is that we choose the separating hyperplane which **maximizes the positive-negative separation margin M** : the greater the distance between the hyperplane and the **closest observed point to this hyperplane** (in the training data), the better.

In other words, rather than choosing *any* set of β values which allows Equation 2 to achieve perfect classification on the training data, we choose the specific set of β values which achieves the maximum margin value M on the training data, so that now the trained classifier in fact satisfies the stricter criterion that

$$\hat{y}_i = \begin{cases} +1 & \text{if } \overbrace{\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}}^{\mathbf{x}_i \text{ at least } M \text{ units "above" hyperplane}} \geq M \\ -1 & \text{if } \overbrace{\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}}^{\mathbf{x}_i \text{ at least } M \text{ units "below" hyperplane}} \leq M \end{cases} \quad (4)$$

perfectly classifies all training datapoints $\mathbf{x}_i$.

(3) Support Vector Classifiers (SVCs)

The final “piece” of the puzzle, that moves us from MMCs to SVCs, is the consideration that we may be able to achieve **better** classification performance¹ if we allow the classifier a certain amount of “slack” in how well it achieves this classification-with-margin- M goal. For example, we might consider providing our classifier with a “**budget**” C determining how many points are “allowed” to violate the strict max-margin classification in Equation 4. Importantly, the SVC optimization is set up such that the **units** of this budget have a particular interpretation:

- With a “budget” of $C = \frac{1}{2}$, the classifier aims to maintain a margin of M overall — meaning, all points are at least M units away from the hyperplane — with the exception that **one** point is allowed to come within $\frac{M}{2}$ of the hyperplane, or **two** points are allowed come within $\frac{M}{4}$ of the hyperplane, and so on. In other words, the classifier must “pay” some amount $c < C$ for each “violation” of the margin by c units
- With a “budget” of $C = 1$, however, the classifier still aims to maintain a margin of M overall, but it now might “spend” this budget to allow one full-on **misclassification** of a point.

In other words, the “units” of the budget C have an important interpretation, such that the difference between $C = \frac{1}{2}$ and $C = 1$ can make the difference between a “perfect” and non-perfect classifier! Each *unit* of the budget corresponds to an “allowed” violation of the margin by a distance of M , and thus any SVC with a budget $C \geq 1$ is “allowed” to misclassify one point, any SVC with a budget $C \geq 2$ is “allowed” to misclassify two points, and so on.

¹Remember that “better” here is relative to our goal throughout the semester, of learning about an **underlying** Data-Generating Process, not just about the particular **realization** of this DGP available to us as training data!

How do we find the “optimal” budget, the budget which achieves the “best” tradeoff between flexibility and overfitting? The answer, as usual lately, is **Cross-Validation!** We can train SVCs using various C values, evaluating each on the basis of their performance on the **validation set**, until we’re satisfied with the level of validation-set performance we’ve achieved (as a “proxy” for how well this particular C value generalizes from seen data to unseen data).

(4) Support Vector Machines (SVMs)

While SVCs are already quite powerful/versatile, they remain dependent on finding “purely linear” separating hyperplanes: the “slack” provided by the budget C only affects *which* line we choose (assuming $p = 2$ for ease of forming a mental picture), but still doesn’t allow the line to “bend”, even if this bending could improve classification performance.

So, just as we moved in earlier weeks from “pure” linear regression models to models like Lasso which allow non-linearity (while still penalizing complexity to avoid overfitting), the move from SVCs to SVMs involves allowing **non-linear** boundaries between the positive and negative subspaces we use to classify points within a p -dimensional feature space.

For intuition-building around this move to non-linearity, the book starts by just referring back to the fact that we can take “purely linear” models like the standard regression model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

and turn them into *non*-linear models by just applying non-linear transformations to the existing features $\{X_1, \dots, X_p\}$ and including these transformed values $\{f(X_1), \dots, f(X_p)\}$ as if they were additional features, retaining the same regression setup:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \beta_{p+1} f(X_1) + \cdots + \beta_{2p} f(X_p)$$

If $f(X) = X^2$, for example, this would enable us to use our original linear regression setup to estimate **quadratic** curves with the same goal of minimizing, e.g., MSE.

The move from SVCs to SVMs *starts* in the same manner — we consider “expanding” the feature space by incorporating nonlinear transformations of feature values as new features within the original SVC setup — but there is a “catch” when it comes to the SVM setup! Namely (as the book struggles mightily to teach in broad strokes, since the underlying math can be a bit of a doozy), SVMs are set up in such a way that they have a “dual representation”:

- In the “primal” representation, used thus far, we learn values for the β parameters, such that each parameter β_j is a multiplier applied to the j th feature value, x_{ij} . That is, to compute a prediction for a given observation $\mathbf{x} = (x_1, \dots, x_p)$, we take each feature value x_j and multiply it by the parameter β_j , summing up these products to obtain a final prediction via Equation 4.
- However, given the mathematical structure of these learned hyperplanes, there is always an equivalent “dual” representation wherein the same hyperplane can be written using a set of parameters α , such that each “dual space” parameter α_i is a multiplier applied to the **inner product** $\langle \mathbf{x}, \mathbf{x}_i \rangle$ between the given observation $\mathbf{x}$ and a training observation $\mathbf{x}_i$.

This “dual representation” at first seems like just a mathematical curiosity, but it has a powerful implication when it comes to SVMs:

- In *general*, “transforming” the problem into the dual space just means estimating n parameters α_1 through α_n (one for each training observation), which may not seem at all preferable to estimating the original $p + 1$ parameters β_0 through β_p ...
- Within the **SVM setup**, however, it turns out that **the parameter values α_i will be non-zero if and only if x_i is a support vector!**

This means that, to make a prediction for a new observation $\mathbf{x}$, our problem is simplified from:

- Computing α_i times the inner product $\langle \mathbf{x}, \mathbf{x}_i \rangle$ for *every* training observation $\mathbf{x}_i$, to
- Computing α_i times the inner product $\langle \mathbf{x}, \mathbf{x}_i \rangle$ *only* for the small subset of “special” observations i^* which happen to be support vectors

In practice, in all honesty, people tend to *use* this SVM “kernel trick” without having a full grasp of the underlying math — in the book, for example, they provide some amount of intuition (which we’ve tried to expand upon a bit in this study guide!), but much of its coverage of SVMs comes by way of the end-of-chapter coding lab. On the quiz, therefore, we will try to emphasize the practical usage of SVMs rather than their mathematical underpinnings, though we still encourage you to try your best to wrap your head around these details!

Practice Problems

Question 1: Separating Hyperplanes as Geometric Objects

Question 1a

Consider a 1-dimensional separating hyperplane defined by the β parameters $(3, 5, -2)$:

$$3 + 5X_1 - 2X_2 = 0 \tag{5}$$

What kind of geometric object does this hyperplane represent?

-
- ☐ A point ☐ A line
☐ A plane ☐ A 3D space
-

Solution:

A line! Since the original $p = 2$ -dimensional feature space is a **plane**, the $(p - 1)$ -dimensional separating hyperplane in this case will be a **line** (as a 1-dimensional geometric object extending infinitely outwards within the plane)

Question 1b

If we want to *visualize* this hyperplane relative to the broader 2-dimensional feature space it is intended to separate, and we plot the first feature X_1 along the x -axis and the second feature X_2 along the y -axis of a 2D plot, what will the **slope** and **intercept** of the resulting line be? Fill in the two blanks below with the appropriate values:

$$X_2 = \underline{\hspace{1cm}} \cdot X_1 + \underline{\hspace{1cm}}$$

Solution:

Solving Equation 5 for X_2 gives us:

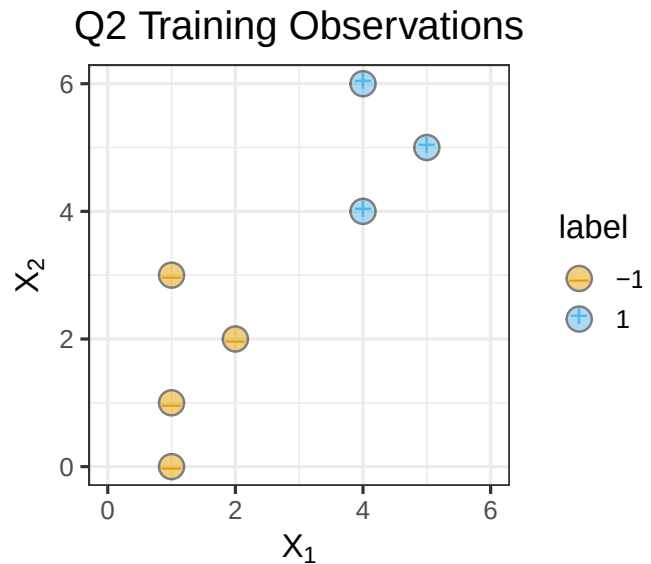
$$2X_2 = 5X_1 + 3 \iff X_2 = \frac{5}{2}X_1 + \frac{3}{2},$$

so that we can fill in the blanks with $\frac{5}{2}$ and $\frac{3}{2}$, respectively.

Question 2: Separating Hyperplanes $\rightarrow$ Max-Margin Classifiers

For each subpart of this question, assume you are trying to construct a binary classifier based on the $N = 7$ training observations given in the table on the left, and plotted on the right:

i	x_{i1}	x_{i2}	y_i
1	4	4	+1
2	2	2	-1
3	5	5	+1
4	4	6	+1
5	1	1	-1
6	1	0	-1
7	1	3	-1



Question 2a

Consider the hyperplane defined by $\beta = (5, 0, -1)$, corresponding to the equation

$$5 + 0 \cdot X_1 - 1 \cdot X_2 = 0$$

Rewrite this hyperplane equation in slope-intercept form, by filling in the following blanks as appropriate, then indicate underneath this equation whether or not this hyperplane is a **separating** hyperplane:

$$X_2 = \underline{\hspace{1cm}} \cdot X_1 + \underline{\hspace{1cm}}$$

Separating hyperplane? ☐ Yes ☐ No

Solution:

We can add X_2 to both sides in the given equation to obtain the desired slope-intercept form as shown on the left side of Figure 1, then we can plot this line as shown in black on the right side of Figure 1.

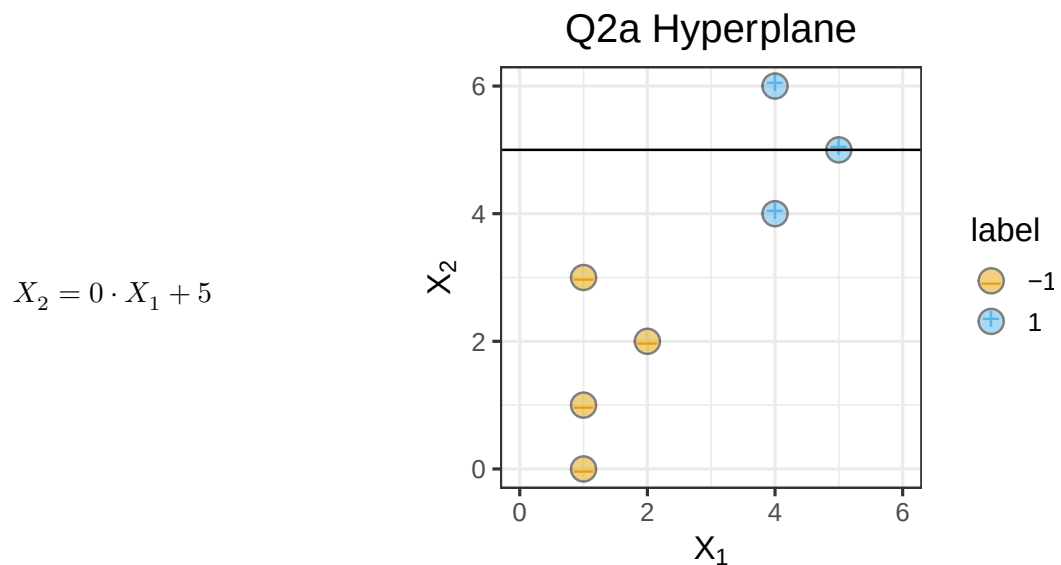


Figure 1: The slope-intercept form and plot of the hyperplane given in Question 2a

From this plot, we see that the hyperplane does **not** separate the positive points from the negative points, and thus it is **not** a separating hyperplane.

Question 2b

Now consider the hyperplane defined *implicitly* as the line

$$X_2 = \left(-\frac{1}{2}\right) X_1 + 2$$

Rewrite this hyperplane in the “standard” hyperplane form $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$, by filling in the following blanks accordingly, then indicate underneath this equation whether or not this hyperplane is a **separating** hyperplane:

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} \cdot X_1 + \underline{\hspace{1cm}} \cdot X_2 = 0$$

Separating hyperplane? ☐ Yes ☐ No

Solution:

In this case, we can re-arrange the equation so that all terms are on one side, to obtain the hyperplane form given in the left side of Figure 2. We can then plot this hyperplane as shown on the right side of Figure 2.

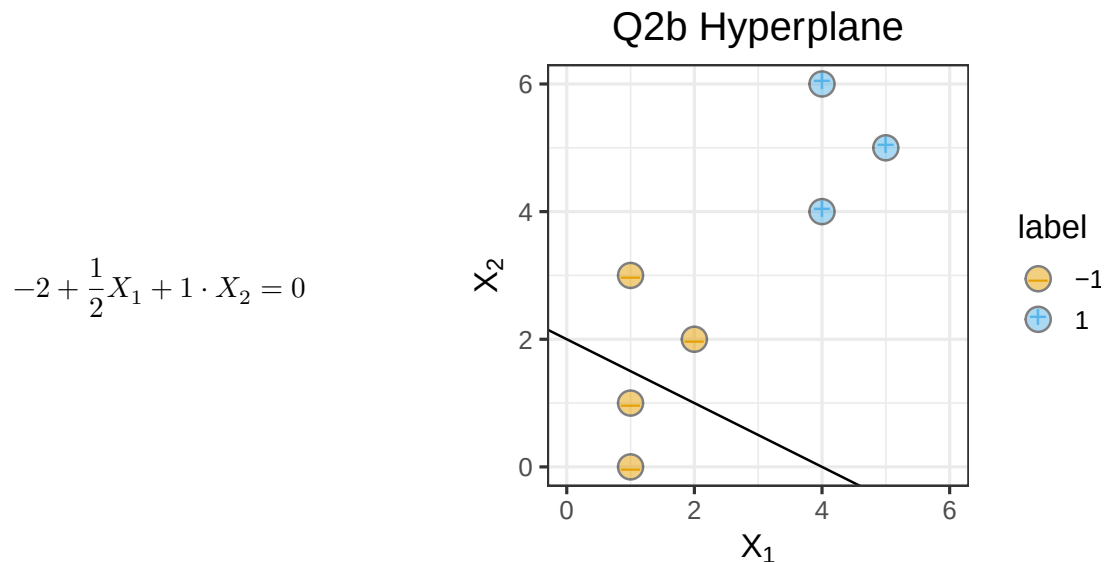


Figure 2: The slope-intercept form and plot of the hyperplane given in Question 2b

And we can see from the plot that, again, we do not have a separating hyperplane.

Question 2c

For each of the following two hyperplanes, indicate whether or not the hyperplane is a **separating** hyperplane. If it *is* a separating hyperplane, compute the **margin** value M achieved by the separating hyperplane and fill in the blank underneath. Otherwise, leave the M value spot blank.

(i)	(ii)
$-2.5 + 1 \cdot X_1 + 0 \cdot X_2 = 0$	$6 - 1 \cdot X_1 - 1 \cdot X_2 = 0$
Separating hyperplane? ☐ Yes ☐ No $M = \underline{\hspace{2cm}}$	Separating hyperplane? ☐ Yes ☐ No $M = \underline{\hspace{2cm}}$

Solution:

In this case *both* hyperplanes are separating hyperplanes, as can be seen in Figure 3.

Hyperplane (i) has a margin value of $M = 0.5$, which you can read directly off of the plot as the horizontal distance between this hyperplane and the negative point at $(2, 2)$ or the positive point at $(3, 6)$.

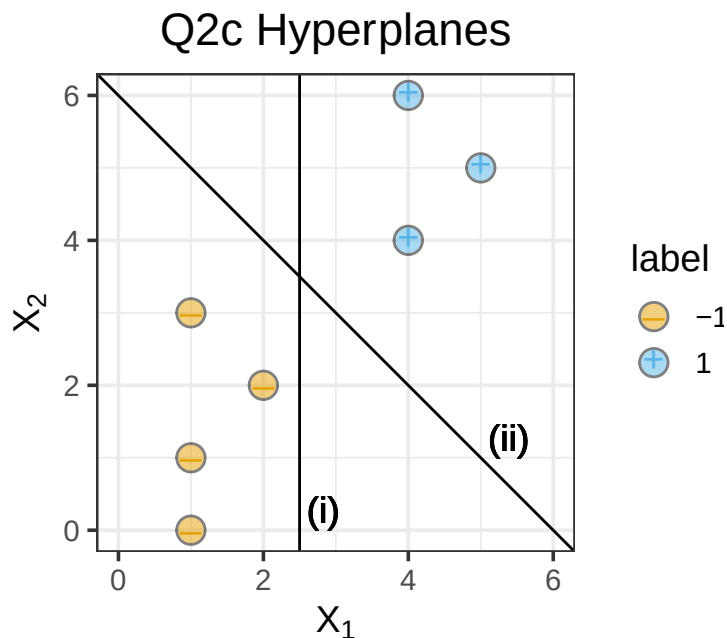


Figure 3: The two hyperplanes given in Question 2c

Hyperplane (ii) has a margin value of $M = \sqrt{2} \approx 1.414$, which you can find by plugging the “center point” of the hyperplane, $(3, 3)$, and the positive point at $(4, 4)$ (or the negative point at $(2, 2)$) into the Euclidean distance formula:

$$M = \sqrt{(4 - 3)^2 + (4 - 3)^2} = \sqrt{1 + 1} = \sqrt{2} \approx 1.414$$

Question 2d

Since the book doesn’t discuss *how* to compute the max-margin hyperplane, for this question you can assume that it is one of the two hyperplanes given in Q2c. Given this information, which of the two hyperplanes is the **max-margin** hyperplane?

Solution:

Since $\sqrt{2} \approx 1.414 > 0.5$, hyperplane (ii) “beats” hyperplane (i) in terms of which achieves a greater margin (and is in fact the max-margin hyperplane!)

Question 3: SVCs

For this question, consider the same training observations as in Question 2, but this time the goal is to learn a Support Vector Classifier with a slack “budget” of C .

Since the book doesn’t go into much detail on the *scaling* that you can apply to the SVC estimate to ensure that this budget C is in *units* of number-of-allowed-misclassifications², for the purposes of

²The relevant detail is mentioned at the bottom of page 372, where it discusses how the constraint (9.10) , $\sum_{j=1}^p \beta_j^2 = 1$, ensures that “the perpendicular distance from the i th observation to the hyperplane is given by $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip})$.”

this question you can just assume that C is set to a value such that you can **ignore the positive point at (4, 4)** in your search for a separating hyperplane — in other words, any amount by which this point violates the margin will simply be absorbed into the budget C ³.

With this new C budget allowing you to ignore the point at (4, 4) in your computation, find the new optimal hyperplane and fill in the blanks below with its β parameters, then indicate underneath this equation the margin value M that this new hyperplane achieves (again, as a hint, you should be able to compute this margin value without the point at (4, 4) playing any role in the calculation!)

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} \cdot X_1 + \underline{\hspace{1cm}} \cdot X_2 = 0$$

$$M = \underline{\hspace{1cm}}$$

Solution:

By examining the plot in Figure 3, but ignoring the point at (4, 4), we can see that a greater margin value M is possible if we place the hyperplane exactly halfway between two downward-sloping lines:

- The line L^+ formed by the two positive points at (4, 6) and (5, 5) and
- The line L^- formed by the two negative points at (1, 3) and (2, 2).

Notice that, since the slope of both L^+ and L^- is -1 , these two lines are parallel. Then, since the distance between these two parallel lines is (taking the first of the two points listed as representative points)

$$\sqrt{(4-1)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2},$$

we can place a hyperplane parallel to these two lines at a distance exactly $M = \frac{3\sqrt{2}}{2}$ from each, to achieve this margin value! We have thus derived one of the two pieces of our answer (the M value).

So, all that's left is to find the equation for a line, parallel to L^+ and L^- , and at this perpendicular distance M away from both.

There *is* a “direct”, general equation for a line parallel to $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$ at a perpendicular distance M from it (that you could derive using e.g. the Pythagorean theorem): $\beta_0 + \beta_1 X_1 + \beta_2 X_2 \pm M\sqrt{\beta_1^2 + \beta_2^2}$.

However, a simpler approach here is to just find a specific point halfway between L^+ and L^- (specifically, we'll find the point halfway between (1, 3) and (4, 6)), then compute the equation for a line passing through this point with the same slope -1 as L^+ and L^- .

First, we compute the point halfway between (1, 3) and (4, 6) as $(\frac{1+4}{2}, \frac{3+6}{2}) = (\frac{5}{2}, \frac{9}{2})$. Then we can compute the intercept of the line with slope -1 passing through this point as

$$\frac{9}{2} = -1 \cdot \frac{5}{2} + b \iff b = \frac{14}{2} = 7.$$

³Given the re-scaling that `scikit-learn`'s `SVC` classifier applies to estimate the SVC hyperplane, for example, choosing a budget value `C` of 1/6 achieves this outcome.

With this intercept value, we can write out the equation for our new separating hyperplane in slope-intercept form:

$$X_2 = -1 \cdot X_1 + 7,$$

which is equivalent to the hyperplane equation

$$7 + (-1)X_1 + (-1)X_2,$$

a hyperplane with coefficients $\beta_0 = 7$, $\beta_1 = -1$, and $\beta_2 = -1$.

Question 4: SVMs

As mentioned in the Review Guide above, the book’s exposition of SVMs mainly focuses on how we can *extend* the SVC model to allow **non-linear** decision boundaries to “chop” the full feature space into positive and negative subspaces.

So, for this question, we will focus on this intuition, whereby SVMs allow us to take a dataset which is **not** linearly-separable (given the “raw” features available for each observation) and “**lift**” the observations into a higher-dimensional space (via non-linear transformations of the original features) wherein they **are** linearly-separable. However, you should keep in mind that this is only *one* of the *two* “core” features of SVMs: the other feature is that SVMs allow us to find separating hyperplanes in this higher-dimensional spaces in **computationally efficient** ways, using the “Kernel Trick”.

Consider the following $N = 3$ observations, where each observation i has only one feature x_i and a label value y_i . The observations are plotted, within their 1-dimensional feature space, on the right.

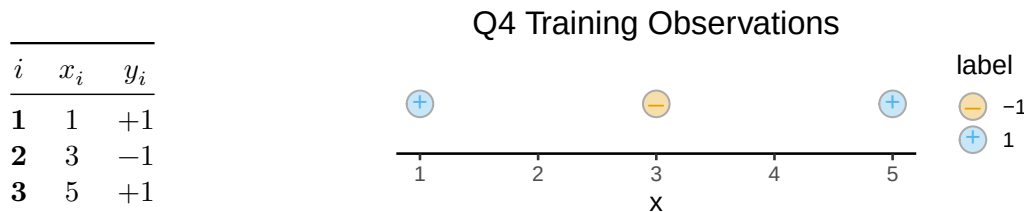


Figure 4: The training observations for Question 4

Question 4a

Are these observations, within their original feature space, linearly separable?

☐ Yes ☐ No

Solution:

No, the training observations are *not* linearly separable within the original feature space.

This is a case where the *general* definition of a hyperplane, as a $(p - 1)$ -dimensional subspace within the original p -dimensional feature space, is helpful to keep in mind: since the feature space is 1-dimensional, a **hyperplane** within this space is actually just a single 0-dimensional **point** in $\mathbb{R}$.

And, since there is no single point we can place on the “numberline” plotted on the right of Figure 4 which would perfectly separate the positive and negative observations (with all positive observations on one side and all negative observations on the other side), there is no hyperplane in this case that would separate the observations as desired.

Question 4b

Find a **transformation** $f(x)$ such that, if we apply this transformation to each feature value x_i and treat the transformed value $f(x_i)$ as a new second feature x_{i2} for each observation, the data becomes linearly separable in the new **2**-dimensional $(x_i, f(x_i))$ feature space.

Once you’ve found this transformation, fill in the blanks below with the coefficients of a **separating hyperplane** which successfully separates the points in the new 2-dimensional space.

$$f(x) = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} \cdot X_1 + \underline{\hspace{2cm}} \cdot X_2 = 0$$

Solution:

There are many possible transformations $f(x)$ that would achieve our goal here, but probably the most straightforward is just $f(x) = (x - 3)^2$. If we apply this transformation to the original data, we get a new set of observations given in the table on the left of Figure 5 and plotted on the right. The question only specified that you should find *some* separating hyperplane, so again lots of possible lines would suffice, but in this case we use the **optimal (max-margin)** separating hyperplane, the horizontal line $y = 2$, which can be written in hyperplane form as

$$2 + 0 \cdot X_1 + (-1) \cdot X_2 = 0$$

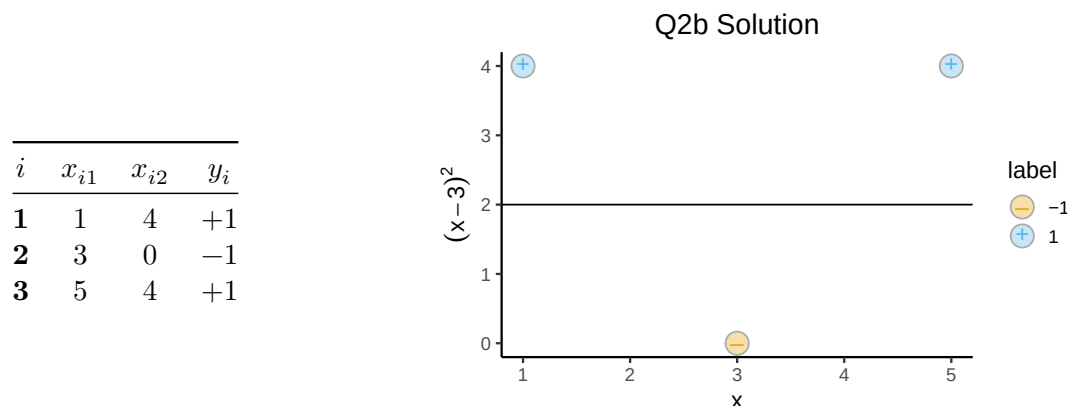


Figure 5: The transformed feature values (x_{i2}) and the plot of the “lifted” 2-dimensional space for Question 4.